

1 • Shared evidence: the research room

The recovery research room

Name _____ Date _____

All numbers are constructed sample data. A, B and C are repeat trials for one fictional person. Rates are beats per minute.

Time	A	B	C
Before activity	72	72	72
Immediately after (0 min)	120	116	124
After 1 min recovery	96	92	100
After 2 min recovery	80	76	84

Source A • NHLBI

During activity, heart rate usually rises to help supply working muscles with more oxygen. This can explain a response; it does not diagnose a person.

Source C • Unattributed advert

“One pulse number tells you exactly who is healthiest.” No author, method or supporting evidence is supplied. This is a fictional claim for evaluation.

Source B • NHS

Regular suitable physical activity can support health. This is broad guidance, not proof that a particular short trial measures total health.

Constructed method

One fictional participant; three separate trials using the same activity, duration, room, recovery posture and measurement schedule. No class trial took place.

W1. Describe the common recovery pattern with two values and units.

W2. Which source best explains heart-rate change? Why? W3. Repair the “healthiest person” headline.

2 • Supported route: one clear briefing

Use trial A and source A on page 1.

S1. What do A, B and C label?

A. Three different people

B. Three repeat trials

S2. Complete: Trial A fell from _____ to _____ beats per minute over two recovery minutes.

S3. Is 80 the same as the starting 72?

A. Yes: fully back to the starting value

B. No: still above the starting value

S4. Complete: NHLBI explains that during activity the heart can beat faster to supply muscles with more _____ .

S5. Write or dictate a 3-sentence briefing: pattern; numbers; one thing the sample cannot prove.

3 • Standard route: evidence-led briefing

Use the whole dataset and source cards.

N1. Describe the common pattern across A-C. Use two pairs of values and units.

N2. Calculate the fall in trial A during the first minute. Show the subtraction.

N3. Choose the most relevant source for the heart-rate explanation. Explain why the advert is weak evidence.

N4. Write a short briefing with claim, evidence, scientific explanation and one limit.

4 • Stretch route: audit the research claim

Keep constructed data separate from real observations and general source information.

T1. Calculate the mean immediate rate and mean rate after 1 minute. Show your working and units.

T2. Compare these means. Does this prove the same fall occurs for every person? Explain.

T3. Identify two controls in the method and one useful improvement. Explain their purposes.

T4. Write a cautious research headline and a 3-sentence briefing. Reject the claim that these data rank three people's health.

5 • Exit and reflection

Give precise evidence, not a diagnosis.

E1. Give one recovery pattern using two values and units.

E2. Name a relevant source organisation and the idea it supports.

E3. Give one limit to what these records can establish.

E4. Name one condition kept the same between trials and explain why.

Reflection. What should the next paper dataset investigate?
